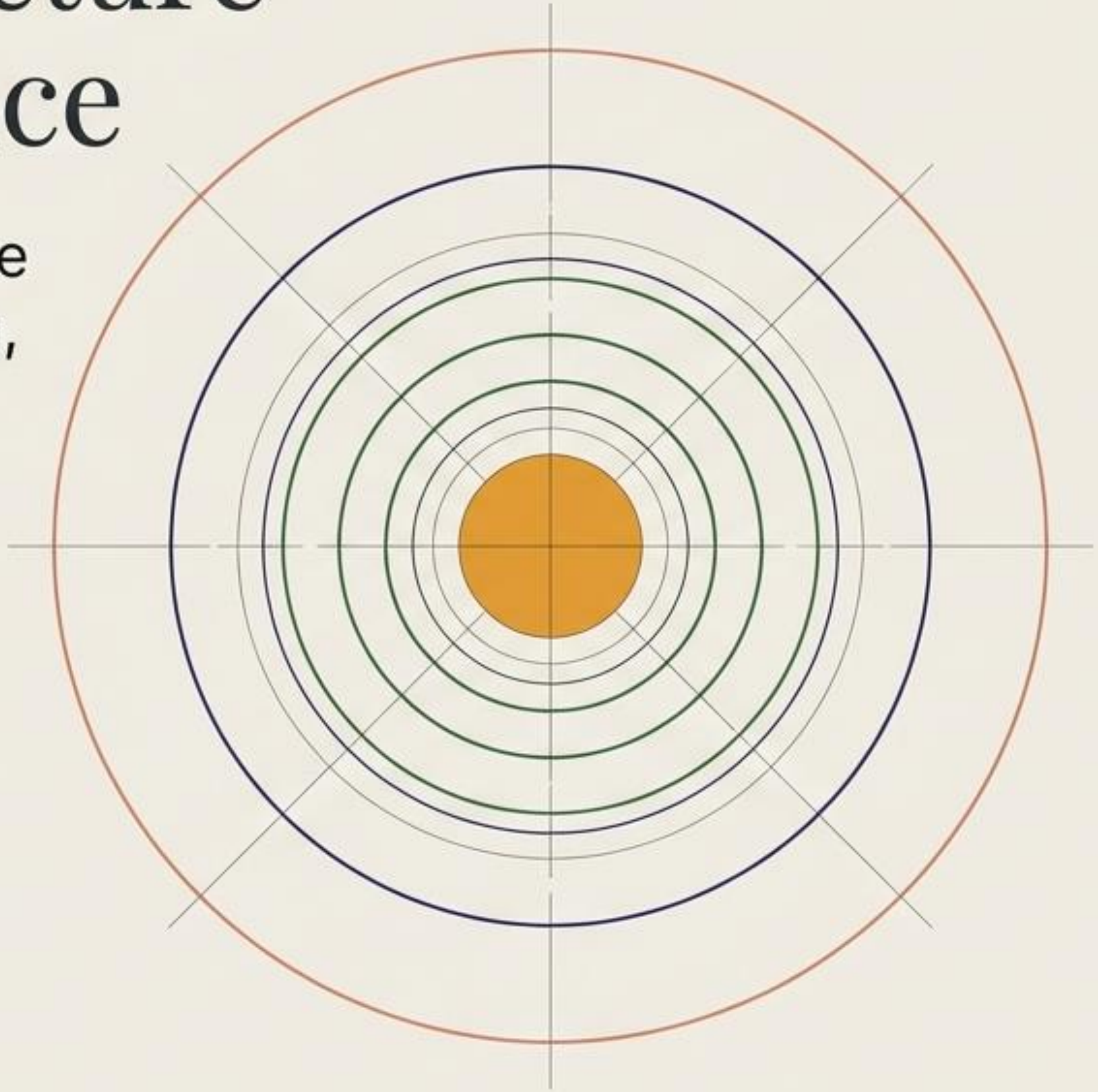
The Architecture of Intelligence

Demystifying AI, Machine Learning, Deep Learning, and Generative AI



Day 1 Curriculum

Mapping the Journey Ahead

1. The AI Ecosystem: Rule-Based Prediction
2. Machine Learning: The Engine of Prediction
3. Deep Learning: Scaling with Neural Networks
4. Generative AI: The Paradigm Shift

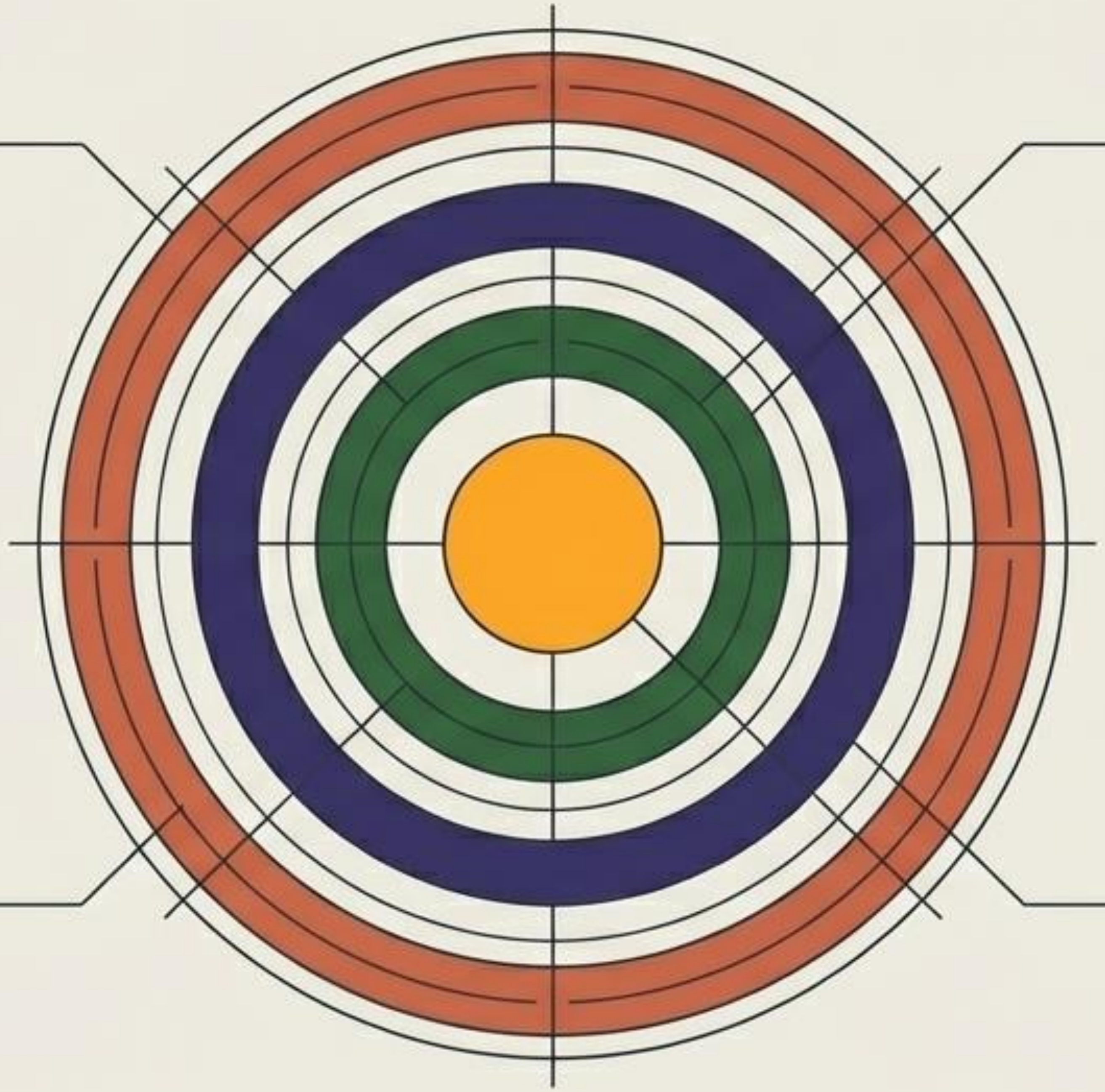
The Blueprint of Artificial Intelligence

Artificial Intelligence (AI)

Machines executing tasks that mimic human behavior and decision-making.

Deep Learning (DL)

A subset of ML utilizing layered neural networks to process vast amounts of data.



Machine Learning (ML)

A subset of AI that learns patterns from data to predict outcomes.

Generative AI (Gen AI)

A subset of DL that shifts from predicting outcomes to generating net-new content.

Artificial Intelligence Mimics Human Decision-Making

Historical & Current Data

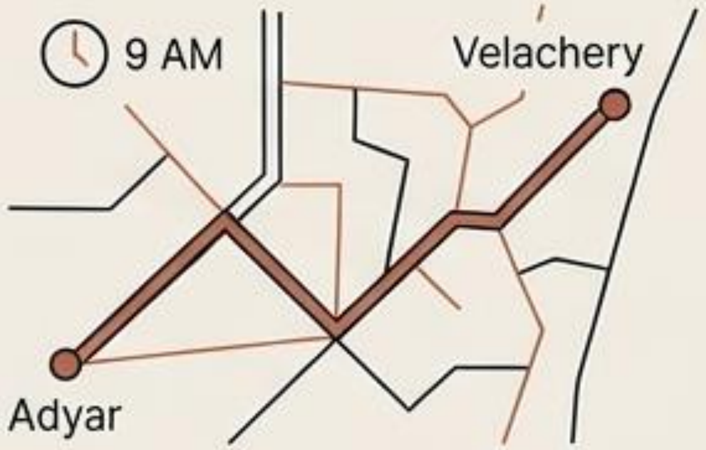
Millions of records processed.

- Current traffic density
- Distance variables
- Historical weather (e.g., 30°C in March across 1990–2024)

The AI Engine

Computing the query:

> "Drive from Adyar to Velachery at 9 AM."



Predictive Output

90-95% Accuracy

On the shortest route and estimated time of arrival.

Machine Learning Powers the Engine of Prediction

A targeted subset of AI that transitions from 1,000s to Millions of records
to refine its predictive accuracy.

'The Scale'

Millions of Records

1,000s

'The Anatomy of Training'

80% Training Data
Learning the historical patterns

20% Testing & Validation
Proving the model's accuracy against unseen data

The absolute output of standard Machine Learning is strictly Predictive.

ML in Action: Multimodal Healthcare Diagnostics

Tabular & Multimodal Inputs

Tabular Data

Age, Blood Group, Diabetic Status, Blood Pressure, Genetic History.

Unseen Multimodal Data

Raw Retinal Images.

The Pipeline

Data Collection → Data Cleansing → Model Selection (CNN: Convolutional Neural Network) → Testing

The Output

Heart Disease Risk: YES / NO

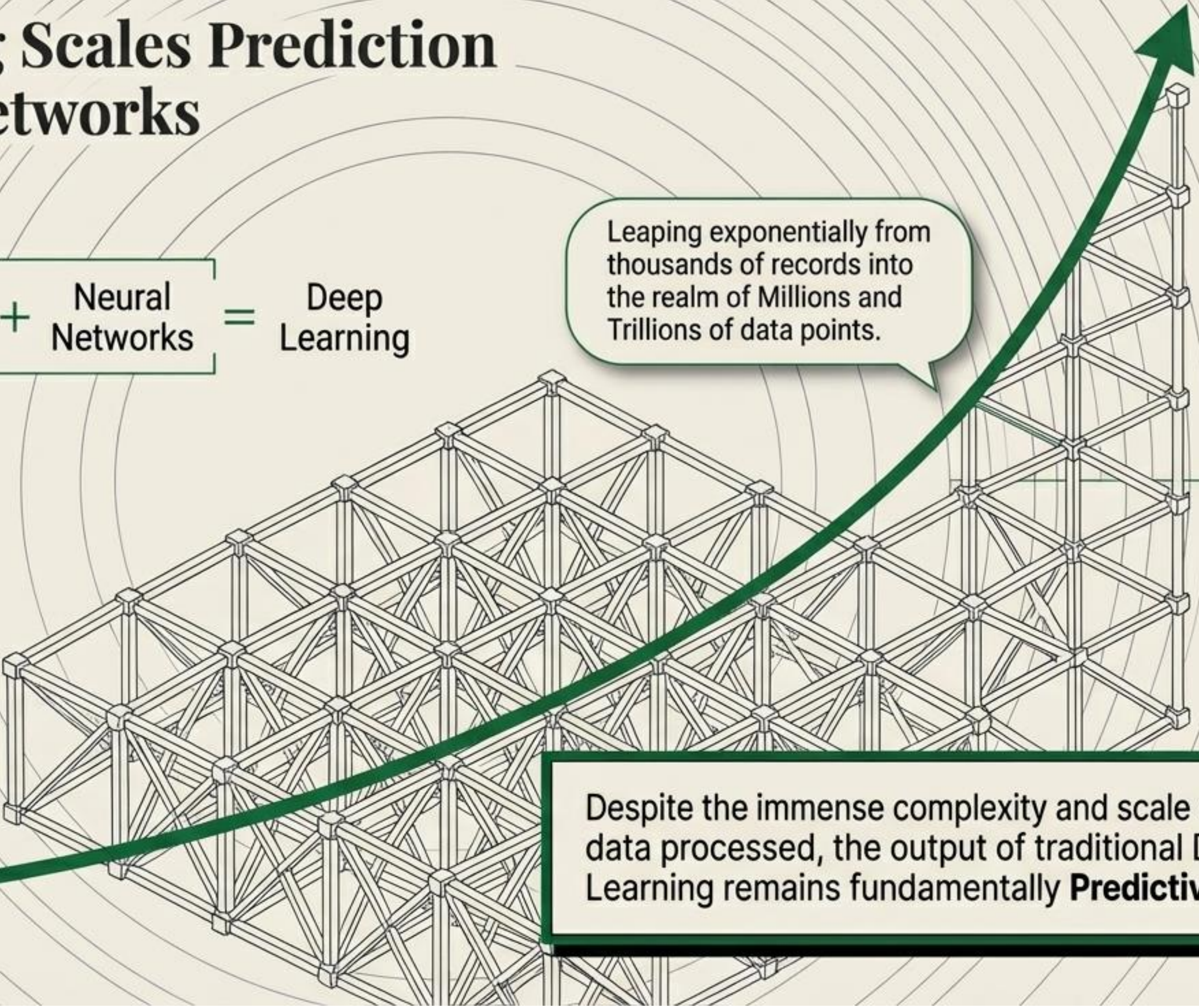
Predictive AI successfully applies learned patterns to unlabelled, unseen retinal images.

The Mechanics of ML: Supervised vs. Unsupervised Learning

Supervised Learning (Labelled Data)	Unsupervised Learning (Unlabelled Data)
Models trained on data with **explicit tags** and **known outcomes** (e.g., Content-based star ratings).	Models that independently **discover hidden structures** and **behaviors** without explicit guidance.
Linear Regression Outputs numerical values (e.g., Home purchase prices, exact defect prediction).	**Behavioral Analysis** Finding unseen patterns in raw data.
Classification Outputs discrete categories (e.g., Heart Attack Risk: Yes or No).	**Collaborative Filtering** Grouping users based on unlabelled behaviors and interactions (without direct likes or comments).

Deep Learning Scales Prediction with Neural Networks

Machine Learning + Vast Data Scale + Neural Networks = Deep Learning

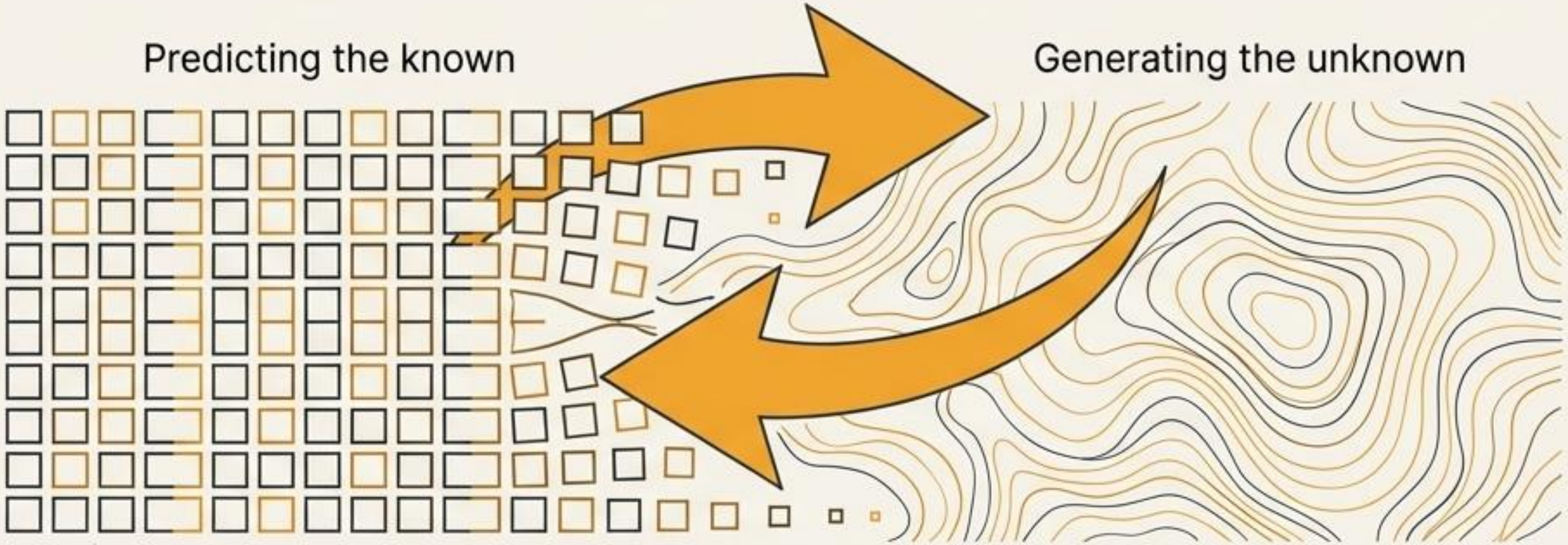


Leaping exponentially from thousands of records into the realm of Millions and Trillions of data points.

Despite the immense complexity and scale of the data processed, the output of traditional Deep Learning remains fundamentally **Predictive.**

Generative AI Represents a Fundamental Paradigm Shift

A specialized subset of Deep Learning that breaks the predictive mold.



The Output Engine: Synthesizing massive datasets to create net-new content.

- Code
- Images
- Text
- Audio
- Video

Large Language Models Power the Generative Core

LLM (Generative Pre-trained Transformer) – Pre-trained on massive, multi-domain datasets, functioning far beyond just software or code generation.

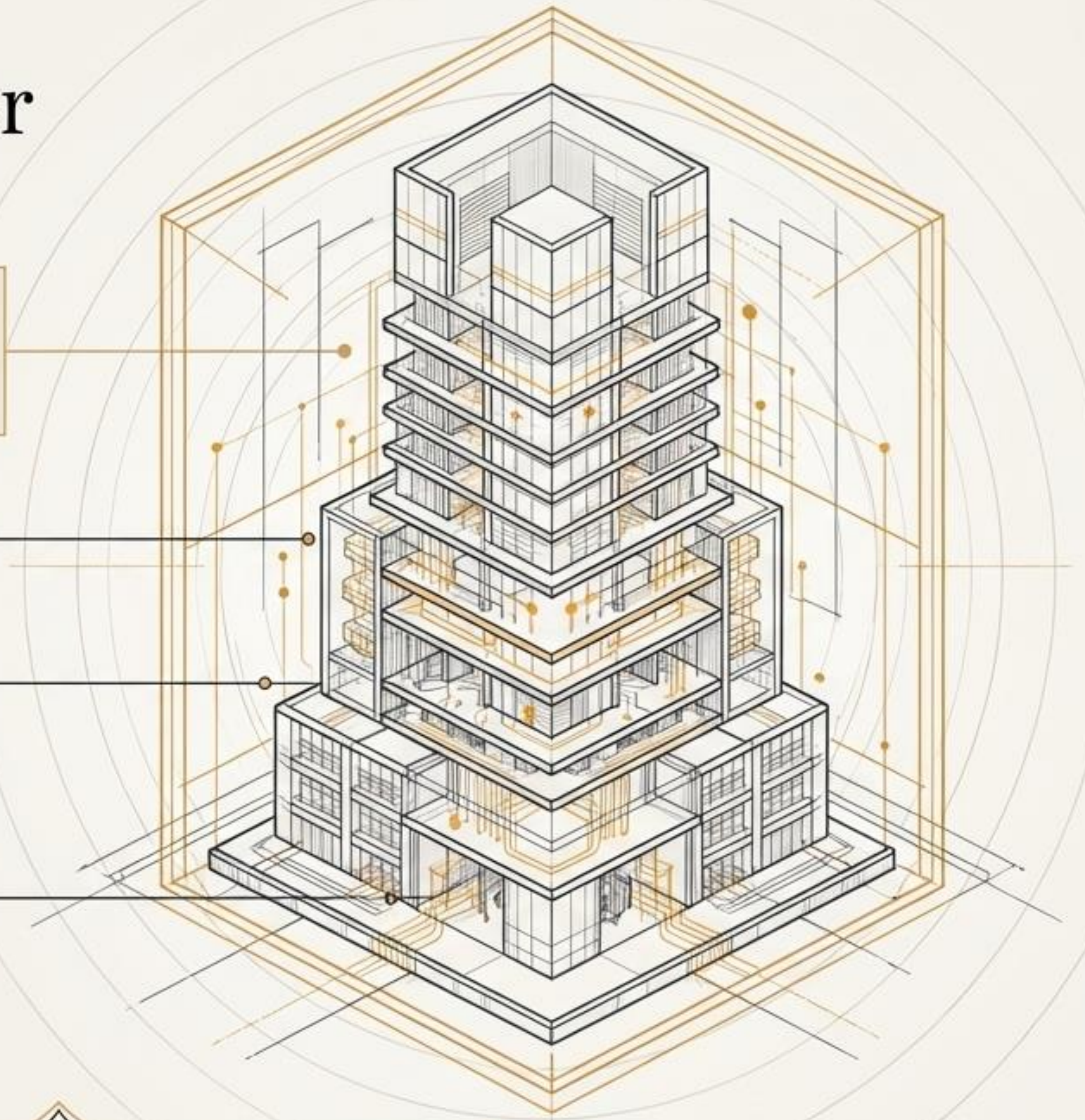
The Architecture of Scale

GPT-4

96 to 120 Architectural Layers.

GPT-5

190 Layers | 120 Billion Parameters.



Real-World Agent: OpenAI's ChatGPT

The Output Divide: Predicting vs. Creating

Predictive AI (AI / ML / DL)

Function: Analyzes historical and current data to guess the future or classify the present.

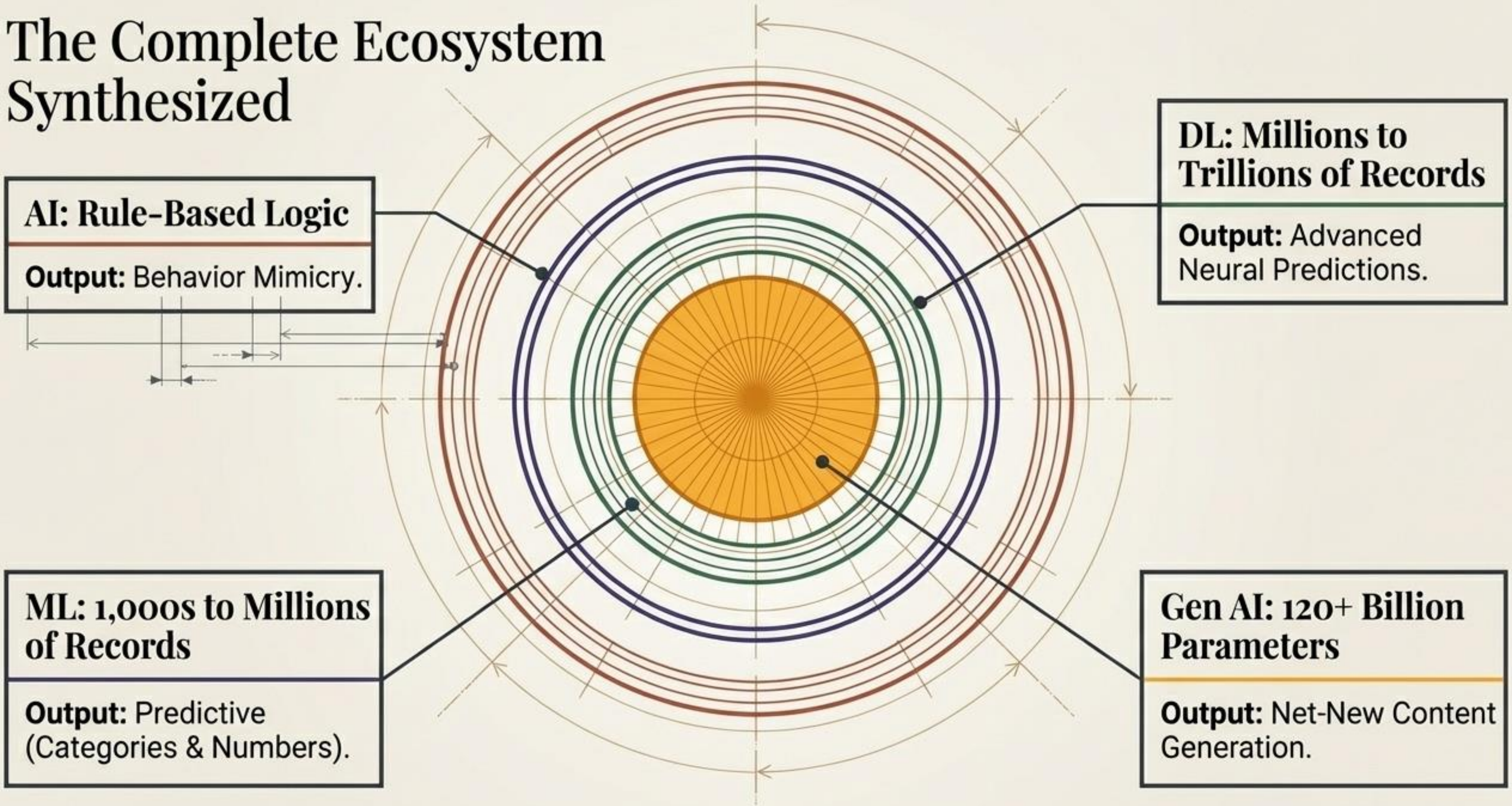
Outputs: Optimal routes, binary risk scores, categories, exact numerical values.

Generative AI (Gen AI)

Function: Understands the deep structural patterns of data to construct entirely new concepts.

Outputs: Original essays, net-new application code, synthesized audio, generated art.

The Complete Ecosystem Synthesized



Preparing for the Deep Dive

- **Today (Day 1):** Mastered the structural evolution, data scales, and the fundamental shift from Predictive to Generative architectures.
- **Next (Day 2):** LLM Anatomy. A complete internal breakdown of the workings, layers, and transformer mechanisms powering modern Generative AI.